Practice Set B

Connections and application | October 2026 revision

75 minutes / 50 marks

Name: _____ Section: _____

Original practice questions, not an official examination or an authentic past-year paper. Answer ALL 30 MCQs in Section A and ALL 5 structured questions in Section B.

Suggested timing: Section A 35 min | Section B 35 min | Check 5 min

Section A answer sheet

Question	Answer	Question	Answer	Question	Answer
1		11		21	
2		12		22	
3		13		23	
4		14		24	
5		15		25	
6		16		26	
7		17		27	
8		18		28	
9		19		29	
10		20		30	

Score: Section A _____ / 30 Section B _____ / 20 Total _____ / 50

How to use this paper

Try it before opening the answer guide. For each mistake, write one corrected idea. Use the course's three-phase model when answering technological-evolution questions.

One topic I need to revisit: _____

Section A | Multiple choice

Questions 1-6 / 1 mark each / Choose ONE best answer.

01. Which scholar-field pairing is accurate?

A. Pythagoras - Arabic medical translation
B. Al-Khwarizmi - mathematics
C. Ibn Sina - steam engines
D. Al-Biruni - internet design

02. A laboratory repeats an observation to check a claim. What feature of science does this demonstrate?

A. Following popularity
B. Checking claims against evidence
C. Choosing conclusions before collecting data
D. Protecting claims from criticism

03. Which action most clearly shows a community meeting fardu kifayah through science?

A. Ensuring enough people have essential medical expertise
B. Treating all occupations as private hobbies
C. Leaving every shared need unmet
D. Requiring every person to become a surgeon

04. A book translator compares a translated passage with its original. Which scholarly role does this illustrate?

A. Preventing cultural exchange
B. Checking the accuracy of translations
C. Replacing scientific study with guessing
D. Limiting all books to one reader

05. A report leaves out failed trials to make a device look reliable. What should the researcher do?

A. Change the failed results into successes
B. Keep only the successful results
C. Report the failed trials accurately
D. Delete the original records

06. Which earlier traditions contributed to the cultural background of Islamic scientific growth?

A. Only twentieth-century digital traditions
B. Only one isolated local tradition
C. Only modern industrial traditions
D. Greek, Persian, Egyptian and Sumerian traditions

Section A | Multiple choice

Questions 7-12 / 1 mark each / Choose ONE best answer.

07. Which statement most clearly confuses fardu kifayah with fardu ain?

A. A community needs enough skilled engineers
B. Scientific expertise may benefit the public
C. Every Muslim must personally qualify as an engineer
D. Different people may serve different community needs

08. A study card names Hunayn ibn Ishaq. Which description belongs beside it?

A. Pioneer of steam-powered manufacturing
B. Scholar associated with the translation movement
C. Author of The Canon of Medicine
D. Greek thinker associated with number theory

09. Why is "Only the final result matters" inadequate in Islamic technological ethics?

A. Useful results never matter
B. Technology should never change
C. The methods used must also satisfy ethical requirements
D. Researchers should avoid all practical goals

10. Which pair correctly matches the roots used in the course?

A. Scientia - craft; techne - knowledge
B. Scientia - wealth; techne - religion
C. Scientia - authority; techne - truth
D. Scientia - knowledge; techne - art or craft

11. Which question most directly reflects an epistemological approach to technology?

A. How do we know that a technical claim is justified?
B. What is the cheapest shipping option?
C. Which advertisement has the brightest colours?
D. How many products were sold yesterday?

12. A library brings together scholars from different religions and backgrounds. Which role does this best support?

A. Keeping knowledge within a single family
B. Ending the need for translation
C. Preventing the training of specialists
D. Exchange of knowledge across communities

Section A | Multiple choice

Questions 13-18 / 1 mark each / Choose ONE best answer.

13. A student labels a medical text "Al-Qanun fi al-Tibb by Ibn Sina." What does the title identify?

A. A manual of steam engines
B. A book of Roman military law
C. The Canon of Medicine
D. A guide to internet networks

14. A researcher notices a pattern and proposes a possible explanation to investigate. What has the researcher formed?

A. A sales forecast unrelated to evidence
B. A hypothesis
C. A rule that cannot be questioned
D. A guaranteed final truth

15. Why can scientific freedom have limits in the Islamic framework?

A. All research questions are prohibited
B. Only wealthy people may investigate
C. Scientific reasoning has no role
D. Researchers have ethical and social responsibilities

16. Which research action is empirical?

A. Measuring and recording water temperature
B. Claiming private inspiration
C. Describing kashf as personal insight
D. Accepting a claim solely through inward experience

17. Which caption best describes Mesopotamian clay tablets containing remedies?

A. Proof of modern vaccination programmes
B. Evidence of digital health records
C. Proof that the internet existed then
D. Evidence of early written medical practice

18. Which statement avoids a common misconception about studying science?

A. Religious responsibility makes observation unnecessary
B. Science can serve both human welfare and religious responsibility
C. Scientific learning has no social value
D. Science is always separate from religion

Section A | Multiple choice

Questions 19-24 / 1 mark each / Choose ONE best answer.

19. A researcher asks whether a conclusion follows from its supporting reasons. Which characteristic is being used?

A. Logic
B. Popularity
C. Blind imitation
D. Status

20. Which contribution fits Al-Biruni in the course outline?

A. Writing The Canon of Medicine
B. Work in astronomy, mathematics and geographical measurement
C. Developing the modern internet
D. Inventing steam-powered factories

21. Which explanation best captures why education matters in the source?

A. It removes the need for truthful reporting
B. It makes ethical judgment unnecessary
C. It helps people overcome ignorance
D. It guarantees that every claim is correct

22. Which summary best describes the link between Baghdad and Baitul Hikmah?

A. A centre designed to stop scholarly exchange
B. A centre established for digital automation
C. A centre of manuscripts, translation and scholarship
D. A centre devoted only to ship construction

23. What does Tawhid contribute to the Islamic view of nature in this course?

A. Belief that scientists have unlimited authority
B. Separation of science from every ethical duty
C. Recognition of the oneness of God
D. Worship of technological progress

24. Which statement about number systems is most accurate?

A. Roman numerals were invented by Al-Khwarizmi
B. Numerals developed without any contact between cultures
C. Islamic scholars helped transmit and develop Hindu-Arabic calculation
D. Algebra means abandoning all number systems

Section A | Multiple choice

Questions 25-30 / 1 mark each / Choose ONE best answer.

25. A project is profitable but unjust to its users. Which judgment best fits Islamic technological principles?

A. Profit does not make injustice acceptable
B. Profit automatically removes ethical concerns
C. Injustice is one of the guiding principles
D. User welfare is unrelated to technological decisions

26. Which phrase best describes the learning attitude encouraged in the early Islamic period?

A. Avoid every unfamiliar idea
B. Replace study with social status
C. Reject all translated texts
D. Seek knowledge widely

27. For the criticism attributed to Dawkins, which classroom response is the best illustration of the concern?

A. "Believe it only because an authority says so."
B. "Can another group repeat the test?"
C. "Let us examine the evidence first."
D. "Show the observations that support it."

28. Which pair of practical developments is associated with Damascus in the source?

A. Internet platforms and electronic records
B. Naval technology and agricultural techniques
C. Steam factories and computer automation
D. Digital computing and satellite networks

29. A student concludes that religion and science must always be enemies. Which course example challenges that conclusion?

A. The meaning of the word techne
B. The existence of scientific observations
C. The recording of medicine on clay tablets
D. The Roman Catholic position summarized as allowing harmony

30. Which example connects scientific study with worship in the Islamic framework?

A. Learning useful science with responsible intentions and conduct
B. Replacing every religious duty with laboratory work
C. Falsifying results to gain prestige
D. Ignoring harm because a project is impressive

Section B | Structured response

20 marks in total / Answer every question. Use the marks to guide the amount of detail.

1. Correct the exhibition labels [4 marks]

Two labels have been switched: Display A shows naval tools and agricultural techniques; Display B shows collected manuscripts and checked translations. Associate EACH display with Damascus or Baghdad, then explain ONE matching contribution.

2. Freedom with responsibility [2 marks]

An engineer says, "Scientific freedom means I can use any method I choose." Give TWO reasons why this is incomplete in the Islamic framework.

3. Explain the functions of Baitul Hikmah [5 marks]

Explain how EACH activity below supported knowledge at Baitul Hikmah: (a) training translators; (b) checking translated texts; (c) involving scholars of different backgrounds; (d) collecting books; (e) encouraging patronage of scholarly work.

Section B | Structured response

20 marks in total / Answer every question. Use the marks to guide the amount of detail.

4. Compare the three technological phases [6 marks]

Using the course model, distinguish the Pre-Revolutionary Phase, Technology Revolution Phase I and Technology Revolution Phase II. For EACH phase, give its main form of production or technology and ONE characteristic or example that makes the distinction clear.

5. Build three accurate study cards [3 marks]

Write THREE scholar-contribution pairs: one about mathematics, one about medicine and one about astronomy or geographical measurement. Use a different Muslim scholar for each pair.